Life Cycle Assessment of Exact 31-mer Counting on an HPC Node: Carbon Payback of an AI-Assisted Optimisation

Isaac Palma-Medina
Master in High Performance Computing
University of Luxembourg
isaac.palma.001@student.uni.lu

Minimum of 4 double-column pages, excluding guidance text, references and appendices.

Compiled October 19, 2026.

***Abstract*—**

I. Introduction

Computation draws electricity, and electricity carries a carbon footprint that depends on where and how it is generated [1]. On a shared high-performance computing (HPC) cluster, a user does not choose the hardware a job draws on: a scheduler, here Slurm, grants an allocation of nodes, cores and memory for a walltime, and the hardware is attributed to the job for that time whether the program uses it or not.

The object of this study is one such job. It counts the exact canonical 31-mers of 5 000 000 sequencing reads of a bacterium, *Mycobacterium abscessus* (FASTA records counted on 19 October 2026), with `kc-py1.py`, a single-threaded Python counter written for teaching [2]. A k-mer is a substring of length k of a DNA sequence. Counting them is a common first step in genome analysis. The job runs on the Aion cluster of the University of Luxembourg, and holds one 128-core node exclusively while its program uses one core.

The footprint of this job is invisible to its user. Slurm returns no energy figure for it (Section III-B), so the user sees a walltime and an exit code, and nothing that shows what the run costs the climate. The allocation shape, more than the program, may decide that cost. An optimisation guided by an AI coding agent also has a cost of its own: the AI assistance and the test runs it requires. It is worth doing only if the savings per run repay that cost within the life of the job.

This paper answers the following research question: *what is the GWP100 carbon footprint of one run of an exact 31-mer counting job that holds an exclusive Aion node, which resource of that allocation carries it, and how many runs does an AI-assisted optimisation of that resource need to pay back its own footprint?* The first two parts concern one particular instance; the third concerns a means [3].

The contribution of this version is the baseline. It is a footprint per run that rests on CPU-package energy counters read by an ordinary user, because the scheduler's own energy fields are empty. It is completed by a declared model of memory power, a proxy power usage effectiveness and an allocated share of hardware production. Each figure is traceable to a measured value, a cited factor or a labelled assumption, and the scripts that recompute them are kept with the job.

Section II positions the study. Section III-A defines the job, the functional unit and the boundary. Section III-B inventories the measured flows. Section III-C converts them into gCO_2e per run, Section III-D interprets that figure, and Section III-E names its hotspots. Section IV attacks those hotspots, and Section V computes the carbon payback. Section VI lists the threats to validity.

II. Related Work

III. Life Cycle Assessment

A. Goal and scope definition

The goal and scope definition fixes what is studied, the functional unit every later figure refers to, the boundary of the system and the method that turns inventory flows into an impact [4], [5].

Goal. This study quantifies the carbon footprint of one execution of a baseline 31-mer counting job. That footprint is the reference against which the AI-assisted optimisation of Section IV and its carbon payback in Section V are measured.

The job. The code is `kc-py1.py` from the public repository `lh3/kmer-cnt` [2], frozen at commit `e2574719cfb784915d80eb5828e78dfae4cfdd7b`. It is a single-threaded Python 3 program. It reads FASTA from standard input and counts every canonical 31-mer, the lexicographically smaller of a 31-mer and its reverse complement. It skips any 31-mer that contains `N`. It prints a histogram with 255 lines, one per count from 1 to 255, and pools every count above 255 into the last line. The input is `M_abscessus_HiSeq_10M.fa.gz` from the repository's v0.1 release [2]: 187 693 955 bytes, SHA-256 `dd2ef49aebabfed1574b55fd002025f29b2fda88c70b2095f5e1b3a822cee736`. It is decompressed on the fly by `gzip`. The program upper-cases only the last record of its input. No sequence line of this input holds a lower-case base (checked on 28 September 2026), so this does not affect the result. The job is submitted with the Slurm script `run-baseline.sh` of the project repository at

commit `253d93ec726b82670c8f4b84da20e9528c9a59ef`. Every run writes the same 1 972-byte histogram, SHA-256 `06a32de77200121a397050e015a257953dc956b7854027a1ed73b15afc12bf42`.

The machine. The job runs on the Aion cluster of the University of Luxembourg HPC platform, partition `batch`, QoS `normal`, on the regular compute node `aion-0127`. An Aion node holds two AMD EPYC 7H12 processors of 64 cores each, 256 GB of DDR4 memory, a 480 GB local SSD and one HDR100 InfiniBand port [6]. The loaded modules are `env/release/default` and `lang/Python/3.11.5-GCCcore-13.2.0` (Python 3.11.5), as recorded in every run's log.

The allocation. Each run holds one node exclusively: one task, 128 cores, all the memory Slurm can reserve on the node (`--mem=0`, recorded as 224 000 MiB) and a walltime limit of 30 minutes. The program uses one of those cores. The node is reserved exclusively so that no other job shares it. As a consequence, the whole node, not one core, is attributed to the job.

The user. The job is submitted with `sbatch` from an Aion login node by the author, through a standard student account.

How many runs. The functional unit is one run. Five runs are measured (Section III-B). The job is assumed to be executed 100 times per year, an assumption listed in Section VI.

Functional unit. The GWP100 carbon footprint of one execution of `kc-py1.py` at commit `e2574719` of `lh3/kmer-cnt`, counting exact canonical 31-mers in `M_abscessus_HiSeq_10M.fa.gz` (187 693 955 bytes, SHA-256 `dd2ef49a...e736`), submitted by a student user on Aion, partition `batch`, holding one exclusive regular node (2 × AMD EPYC 7H12, 128 cores, 224 000 MiB reservable memory) for a median walltime of 368 s.

Method. One criterion: a carbon footprint in CO_2e. The characterisation method is IPCC 2021 [7] and the indicator is GWP100.

Boundary. Inside the boundary are the node the job holds for its walltime, and the share of the interconnect, the storage and the site that serve it. The last three enter through the power usage effectiveness (PUE) declared in Section III-B. Outside the boundary are the author's laptop, the login node, the network between laptop and cluster, the one-off download of the input, the development of the code, and every node and job the allocation does not reserve.

Life cycle phases. For the compute node, the study keeps production, including the extraction of raw materials that cradle-to-gate data contain, and use. For the interconnect, the storage and the site, it keeps use only, through the declared PUE. End of life is not inventoried separately: the disposal route of Aion's hardware is not published. One of the candidate hardware datasets of Section III-B nonetheless includes disposal. That mismatch is listed in Section VI.

Components. The study is capped at four components, which Section III-B inventories in this order: (1) the *compute node*: its electricity during the walltime and a share of its manufacture; (2) the *interconnect*: the HDR100 InfiniBand fabric that carries the job's reads and writes between the node and shared storage; (3) the *storage*: the shared GPFS file system that holds the input, the code and the output; (4) the *site*: cooling and power distribution around the machine.

B. Life cycle inventory

The inventory lists the physical flows the assessment runs on, and how each was obtained. It holds no grams: those belong to Section III-C. Every entry is gathered in Table I in Appendix A, with its source and the date it was read. The raw scheduler records, logs and commands are kept in the project repository.

The measurement. The frozen commit and input of Section III-A were run five times as separate batch jobs on 29 September 2026, submitted consecutively with `--dependency=singleton` so that no two overlapped. Each run held its node exclusively, and all five landed on `aion-0127`. All five completed with exit code 0 and wrote identical output. All five are kept and the median is reported. Walltimes were 362, 364, 368, 369 and 372 s: a median of 368 s and a spread of 10 s from fastest to slowest. File system caches were neither reset nor measured. The first run is not the slowest (the third is), so these runs show no first-run effect, but they do not establish whether any run started cold. The records were read with `sacct` on 29 September 2026. The fields read, and why each is the quantity claimed, are: `Elapsed`, the time from start to end of the allocation, which is the walltime the node is held; `AllocNodes` and `AllocCPUS`, the node and cores held; `ReqMem`, the memory reserved; `MaxRSS` of the Python step, the peak resident memory the program touched, which agrees with the maximum resident set size that `/usr/bin/time -v` reports to within 0.2 %; `MaxDiskRead` and `MaxDiskWrite` of the batch and Python steps; and `ConsumedEnergy` and `ConsumedEnergyRaw`.

Whether an energy figure comes back. It does not come back from the scheduler. On 29 September 2026, `scontrol show config` reported the energy plugin `acct_gather_energy/rapl`. Nevertheless, both energy fields were empty for every step of all five jobs, and for a separate interactive allocation checked the same day. The empty fields are missing data, not zero energy. The processor's own RAPL counters are readable by an ordinary user through `/sys/class/powercap`, while `perf` reports its package energy event as not supported. The job was therefore run five more times on 29 September 2026, with the same program, input and allocation. Each run read both package counters directly before and after the Python step, and corrected for counter wrap. `aion-0127` was busy, so these five runs and an idle reading were moved to `aion-0144`, an identical regular node. All five completed with exit code 0 and wrote the same histogram, and Slurm's energy fields stayed empty. Over Python steps of 357.6 to 366.4 s, the two CPU packages drew a median of 67 834 J (range 67 129 to 68 827 J), an average of 188 W. Read the same way for 60 s after the fifth run, the idle node drew 10 959 J, or 183 W. Assuming that idle rate

held throughout each run, the job's activity adds a median of only 1 863 J (range 1 690 to 1 953 J). So 97 % of the package energy is drawn by a node the job holds but barely uses. RAPL covers the processor packages only: memory, board, local disk, fans and network card are outside it, and so is the time the allocation runs outside the Python step. Because no instrument reaches the whole node, the node's electricity is also estimated with the Green Algorithms model [1]: the walltime times the power of the cores and memory held, using 280 W per 64-core processor [6] and 0.3725 W per GB of memory. The whole 256 GB installed is counted, because the exclusive allocation holds all of it. Counting one active core gives 36.7 kJ (0.0102 kWh) per run. Counting all 128 allocated cores at full power gives 241 kJ (0.0670 kWh). The measured package energy alone, 67.8 kJ, is almost twice the first figure. So the one-core model understates the node, and both model figures are scenarios, not bounds. The measured package energy is the figure this inventory carries forward.

Allocation rule. Anything shared across many runs is divided down to one run by the node-seconds the job holds over the node-seconds the hardware delivers during its service life. For one exclusive node this is the walltime over the service life. The same rule is applied in every subsection below.

1) Compute node: Operational: 67 834 J of CPU-package electricity per run (median, RAPL), of which 1 863 J is above idle, over a median walltime of 368 s, holding 1 node and 128 cores. Memory reserved: 224 000 MiB (`ReqMem`), of the 256 GB installed [6]. Memory touched: 7 760 436 KiB, or 7.40 GiB (median `MaxRSS`). The job therefore touches 3.4 % of what it reserves, and 211 GiB stays reserved but unused. Embodied: ecoinvent cut-off v3.11 [8] holds no server production activity. Searches on 29 September 2026 for *server* (58 activities, 32 products), *computer production*, *computer, desktop* and *computer, laptop* found none. The closest dataset is *computer production, desktop, without screen* (GLO, 1998–2024, per unit). It models a 2002 single-processor desktop of 11.3 kg and includes packaging and disposal. Under IPCC 2021 GWP100 it carries 221 kg CO_2e per unit. A two-socket AMD EPYC server, the Dell PowerEdge R7525, has a published production footprint of 1 768.4 kg CO_2e, of which 854.7 kg comes from two 4 TB SSDs [9]. That study uses IPCC 2013 rather than IPCC 2021, and its server differs from an Aion blade. Neither figure describes an Aion node. Which one enters Section III-C, and why, is decided there. Aion entered production in October–November 2021 and still runs in September 2026 [6]. No retirement date is published, so a service life of six years is assumed (Section VI). One run then holds 368 s of 189 345 600 node-seconds, a share of 1.94×10^{-6} of the node.

2) Interconnect: The job runs on one node, so no node-to-node traffic occurs. Its only traffic on the HDR100 fabric (link rate 100, read with `ibstat`) is file system I/O. The step counters do not measure storage traffic directly. The Python step's `MaxDiskRead` of 680.67 MiB equals the decompressed stream that `gzip` writes into the pipe (713 674 148 bytes, or 680.61 MiB). The batch step's `MaxDiskWrite` of 680.72 MiB is `gzip` writing that same pipe. What the batch step reads beyond the pipe, about 198 MiB, is the 179 MiB compressed input plus about 19 MiB of other files, most likely the loaded modules. The data read from storage per run is therefore about 187 693 955 bytes plus about 19 MiB, and the data written is the 1 972-byte histogram plus the job log. This breakdown is inferred from the counters, not measured on the fabric. Runs 2 to 5 may have read the input from the node's page cache. The fabric's energy is not metered per job. It enters through the declared PUE of the site subsection.

3) Storage: The repository, input and output sit on Aion's shared GPFS file system (`df -T`, 29 September 2026) [6]. Per run, the job keeps the 187 693 955-byte input and writes the 1 972-byte output, reading and writing the bytes listed under the interconnect. Storage energy is not metered per job. It enters through the declared PUE of the site subsection.

4) Site: The University of Luxembourg HPC centre publishes no PUE for Aion. This study therefore declares 1.36, the EU average PUE for 2023, reported in 2024 and published in the European Commission's first technical report on data-centre energy performance [10], read on 29 September 2026. It is a regional proxy, not a measurement of this site. It multiplies the operational energy of the compute node, and covers cooling, power distribution, interconnect and storage. How far a different value would move the result is discussed in Section VI.

C. Life cycle impact assessment

The impact assessment multiplies each inventory flow by a characterisation factor, an impact per unit of flow, chosen, cited and motivated. This study assesses one impact category, climate change. It is a single-criterion carbon footprint in CO_2e. It leaves out water use, mineral resource use and every other category of the Environmental Footprint method. The characterisation method is IPCC 2021 [7], and every factor below is read as GWP100. The chain from joules to grams is gathered in Table II in Appendix A. Every value in it is recomputed from the raw records by the script `lcia-calculation.py` of the project repository.

The energy boundary. The measured CPU-package energy of Section III-B, 67 834 J, covers the two processor packages during the Python step only. The job, however, holds the whole node for its walltime. This study therefore builds a node estimate from three terms, each taken per run from the five `aion-0144` runs and paired with those runs' own timing: (1) the measured package energy over the Python step; (2) the idle package rate, 182.6 W, over the rest of the allocation, the gap between Slurm's `Elapsed` and the Python step (1 to 3 s per run, 0.2 to 0.6 kJ); (3) the memory, modelled with the Green Algorithms coefficient of 0.3725 W/GB [1] for the 256 GB installed, 95.4 W, over `Elapsed`. The median node estimate is 103 146 J (range 101 622 to 104 215 J), or 0.0287 kWh, over a median `Elapsed` of 364 s. It is still partial. The board, the local SSD, the network card and power conversion are outside it, and Section VI gives their likely size. The functional unit's 368 s comes from the `aion-0127`

series, which carries no energy reading. The energy chain uses the 364 s of the `aion-0144` runs instead, so that energy and time come from the same runs. The two medians differ by 1.1 %.

Site. The declared PUE of 1.36 [10] (Section III-B) multiplies the node estimate: 0.0287 kWh $\times$ 1.36 $=$ 0.0390 kWh drawn by the site per run.

Grid factor. Aion is in Luxembourg. The factor is ecoinvent 3.11 cut-off, *market for electricity, low voltage*, LU, time period 2020–2024, IPCC 2021 climate change total excluding biogenic CO_2, GWP100 [11]: 337 gCO_2e/kWh, read on 19 October 2026. It describes the electricity consumed in Luxembourg, imports and grid losses included, which is the electricity the site draws. A factor that counts only generation on national territory would leave out the imports. Operational impact: 0.0390 kWh $\times$ 337 g/kWh $=$ 13.1 gCO_2e per run.

Hardware production. Section III-B holds two proxies. This study uses the Dell PowerEdge R7525 [9], a two-socket AMD EPYC server, because the ecoinvent desktop dataset models a 2002 single-processor machine of 11.3 kg. The R7525 figure is 1 768.4 kg CO_2e, of which 854.7 kg comes from two 4 TB SSDs. An Aion node holds one 480 GB SSD, so the two SSDs are removed, which leaves 913.7 kg. The Dell study characterises with IPCC 2013, not IPCC 2021, and publishes no inventory per gas, so the figure cannot be recharacterised. For CO_2 the two methods agree, with a GWP100 of 1 by definition. Between them, the GWP100 of fossil methane moves from 28 to 29.8 and that of N_2O from 265 to 273 [12], [7], which are changes of 6 % and 3 %. A change of that order on the hardware term moves the total by less than 1 %. The figure is therefore kept and labelled IPCC 2013 in Table II. The grid factor is not applied to it, because the hardware is made worldwide. With the allocation rule of Section III-B and the 364 s of the paired series, one run holds $364/189\,345\,600 = 1.92 \times 10^{-6}$ of a six-year life, so 913.7 kg $\times$ 1.92×10^{-6} $=$ 1.76 gCO_2e per run.

Result. One run of the baseline job emits $13.1 + 1.8 \approx 15$ gCO_2e, to two significant figures.

D. Interpretation

The figure of 15 gCO_2e is the GWP100 footprint of one run of the frozen job holding one exclusive Aion node for 364 s. It counts the measured CPU packages, a modelled memory, the site through a proxy PUE, and a time share of a proxy server's production.

Against daily life. A 5 W LED bulb lit for one hour draws 5 Wh, or 1.7 gCO_2e on the same Luxembourg factor. One run therefore equals about 9 hours of that bulb. It also equals about 50 m in a small petrol car, at 291 gCO_2e per km [13]. That figure is cradle to grave and includes the road, so it has a wider boundary than the footprint of one run. At the assumed 100 runs per year, the job emits 1.5 $kgCO_2e$ per year.

Against the alternative scenarios. Table III in Appendix A puts the chosen estimate next to the energy scenarios of Section III-B, all on the 364 s of the paired series. The Green Algorithms figures of Section III-B, 36.7 and 241 kJ, used the 368 s of `aion-0127`, so they are 1.1 % higher than the 36.3 and 238.6 kJ shown here. Each gap has one cause. The package energy alone, 67.8 kJ, lacks the memory term (34.7 kJ) and, for the median run, the 0.6 kJ outside the Python step. It is 34 % below the chosen figure. The one-core model, 36.3 kJ, counts 4.4 W of processor power where the packages measure 188 W, because it gives the 127 unused cores no power at all. That accounts for its entire 66.8 kJ gap, since the memory term is the same in both. The all-cores model, 238.6 kJ, counts 560 W of processor power, the rated maximum of both packages. That is 135 kJ more than the chosen figure. The package energy above idle, 1.9 kJ, is the work the program itself adds. It is less than 2 % of the chosen figure.

Which check this is. The all-cores model is a top-down estimate: the node held, times its rated processor power, times the walltime. Next to a counter reading, it tests the reading. The measured 188 W is 34 % of the rated 560 W. That ratio is plausible for a node on which one core of 128 works, so the check supports the reading and does not contradict it. The one-core model is a substitute. Next to the reading, the substitute is the figure under test: used on an exclusive node, it understates the processor power 43-fold (4.4 against 188 W) and the node estimate almost threefold. The chosen result keeps the reading, and uses a model only for the memory, which no counter reaches.

E. Reporting

The baseline is 15 gCO_2e per run: 13.1 g of electricity and 1.8 g of hardware production. The functional unit is one run of `kc-py1.py` on the fixed input, holding one exclusive regular Aion node on partition `batch` for 364 s. The conventions are IPCC 2021 GWP100, CPU packages measured with RAPL over the Python step, memory modelled at 0.3725 W/GB, a PUE of 1.36, a grid factor of 337 gCO_2e/kWh and a six-year hardware life. Across the node scenarios of Table III, the total ranges from 6.4 to 32 gCO_2e. A reader who reuses the figure should scale it with the node-time held, not with the CPU work done.

Two hotspots carry the result, and both come from the allocation shape rather than from the counting. The first is the idle draw of the CPU packages of the exclusive node: 8.5 gCO_2e, or 57 % of the total with its share of the site overhead. The program's own work above idle takes 1.6 %. The second is the reserved memory: 4.4 gCO_2e, or 30 %. The job touches 3.4 % of that memory. The remaining 12 % is hardware production. Section IV attacks these two hotspots: one 128-core node, held whole, for a program that uses one core and 7.4 GiB.

IV. Optimisation

A. Optimisation procedure

B. Implementation

C. Footprint after optimisation

D. What generalises

V. Carbon Payback

A. What this means

VI. Threats to Validity

This section lists the limitations of Section III-C that remain open. For each, it gives the direction in which it moves the baseline of 15 gCO_2e and, where possible, by how much. Each shift below moves one input of `lcia-calculation.py` and holds the others fixed.

Missing whole-node power. No instrument reaches the whole node. RAPL covers the processor packages only, and the memory term is a generic coefficient [1], not a reading. The board, the local SSD, the network card and power conversion are not counted, so the result is too low. Their power on an Aion node is not published. Each 50 W they draw adds 2.3 gCO_2e per run, or 16 %. The memory coefficient can err in either direction. The idle rate assumed outside the Python step concerns at most 0.6 kJ per run, under 1 % of the node estimate. The idle reading is one 60-second interval taken after the fifth run. If the idle draw varies over time, the 57 % hotspot share moves with it.

Proxy PUE. The PUE of 1.36 is an EU average for 2023 [10], not a value measured at this site. Aion's blades are built on a cold plate that cools every component by direct contact [6], which may put the site below the average, but no site value is published. A PUE of 1.1 gives 12.4 gCO_2e, 17 % less, and 1.6 gives 17.2 gCO_2e, 16 % more. Each step of 0.1 moves the result by about 1.0 g.

Grid factor. The factor describes low-voltage consumption, including distribution losses. If the site is supplied at a higher voltage, part of those losses does not apply, and the result is slightly too high. The voltage of the connection is not published. The factor rests on the 2020–2024 market and is shown to three significant figures. Its documentation also refers to the high voltage level, against its own title.

Proxy hardware data. No production footprint exists for an Aion node. The R7525 is a rack server with its own chassis, fans and power supplies. An Aion node is one of three nodes in a cold-plate-cooled blade [6]. The SSDs are removed on the assumption that the rest of the R7525 resembles the node. With the ecoinvent desktop instead, the hardware term falls to 0.43 g and the total to 13.6 g, 9 % less. With the R7525 as published, the hardware term rises to 3.4 g and the total to 16.5 g, 11 % more. The R7525 figure uses IPCC 2013. For methane and N_2O this moves the hardware term by a few per cent and the total by less than 1 % (Section III-C). The fluorinated gases of chip making are not broken down in the Dell study, so their change between the two methods is not checked. The production of the chassis, the switches, the storage servers and the building is left out, so the result is too low.

Assumed service life. The six-year life is an assumption. Aion has run for five years and has no published retirement date [6]. A four-year life gives 15.8 g, 6 % more, and an eight-year life gives 14.4 g, 3 % less. The allocation also assumes that the node is available for every second of its life. Downtime and idle periods between jobs spread production over fewer node-seconds, so the hardware term is too low by that unknown fraction.

Cache state. File system caches were neither reset nor measured, so a run may have read its input from the page cache rather than from GPFS. The Python step is bound by computation: its CPU time is 99.7 % of its elapsed time in job 15861652. Reading from storage is therefore a small part of the walltime. The walltimes of the five paired runs span 359 to 368 s, and the node estimate spans 101.6 to 104.2 kJ. A cache effect is bounded by that 2.5 % spread.

Measurement on different nodes. The walltimes of the functional unit come from `aion-0127`, and the energy from `aion-0144`. The chain uses only the `aion-0144` values. With the 368 s of `aion-0127`, the result would be 15.05 instead of 14.89 gCO_2e, 1.1 % more. An idle package reading on a third node, `aion-0234`, gave 178.6 W against 182.6 W on `aion-0144`. Identical nodes thus differ by about 2 % at idle.

Figures with different boundaries. The electricity terms cover use only. The R7525 figure covers production, while the ecoinvent desktop includes packaging and disposal. End of life is outside the result, so the result is too low by the unknown share of disposal. The car comparator of Section III-D is cradle to grave, which makes one run look smaller next to it than a comparison on the same boundary would.

Runs per year. The figure of 100 runs per year is an assumption. It leaves the result per run unchanged and scales only the annual figure and the payback period of Section V.

One criterion, burden shifting. The result covers climate change only, on one job. It hides the mineral and water use of hardware production. A change that frees the node may also lead the job to be run more often. The result supports no comparison with another job.

AI Usage Transparency

References

[1] L. Lannelongue, J. Grealey, and M. Inouye, "Green Algorithms: Quantifying the carbon footprint of computation," *Advanced Science*, vol. 8, no. 12, p. 2100707, 2021.

[2] H. Li, "kmer-cnt: Code examples of fast and simple k-mer counters for tutorial purposes," https://github.com/lh3/kmer-cnt/tree/e2574719cfb784915d80eb5828e78dfae4cfdd7b, 2020, commit e2574719cfb784915d80eb5828e78dfae4cfdd7b, MIT licence. Accessed 28 September 2026.

[3] M. Shaw, "Writing good software engineering research papers," in *Proceedings of the 25th International Conference on Software Engineering (ICSE 2003)*. IEEE Computer Society, 2003, pp. 726–736.

[4] *ISO 14040:2006, Environmental management: Life cycle assessment: Principles and framework*, International Organization for Standardization Std., 2006.

[5] *ISO 14044:2006, Environmental management: Life cycle assessment: Requirements and guidelines*, International Organization for Standardization Std., 2006.

[6] ULHPC Team, "Aion: compute nodes, interconnect, timeline, and GPFS file system," ULHPC Technical Documentation, https://hpc-docs.uni.lu/systems/aion/compute/, https://hpc-docs.uni.lu/systems/aion/timeline/, https://hpc-docs.uni.lu/filesystems/gpfs/, 2026, University of Luxembourg. Accessed 29 September 2026.

[7] P. Forster, T. Storelvmo, K. Armour, W. Collins, J.-L. Dufresne, D. Frame, D. J. Lunt, T. Mauritsen, M. D. Palmer, M. Watanabe, M. Wild, and H. Zhang, "The Earth's energy budget, climate feedbacks, and climate sensitivity," in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2021, pp. 923–1054.

[8] ecoinvent Association, "ecoinvent database version 3.11, system model allocation, cut-off by classification," ecoQuery, https://ecoquery.ecoinvent.org/3.11/cutoff/dataset/3193/impact_assessment, 2024, dataset: computer production, desktop, without screen (GLO). Released 20 November 2024. Accessed 29 September 2026.

[9] Sphera Solutions, "Life cycle assessment – Dell servers R6515, R7515, R6525, R7525," Prepared on behalf of Dell Technologies, Report v1.0, Feb. 2021, table 4-5, p. 47. Accessed 29 September 2026. [Online]. Available: https://www.delltechnologies.com/asset/en-us/products/servers/technical-support/full-lca-of-dell-severs-r6515-r7515-r6525-r7525.pdf

[10] F. Heatubun, D. Dragan, M. Bolchi, and M. Lopes Bautista, "Assessment of the energy performance and sustainability of data centres in EU: First technical report," European Commission, Directorate-General for Energy; Publications Office of the European Union, Luxembourg, Tech. Rep., Jul. 2025, prepared by EY with Borderstep and AIT. Figure 6, p. 34: EU average PUE 1.36 for 2023 data. Accessed 29 September 2026.

[11] ecoinvent Association, "ecoinvent database version 3.11, system model allocation, cut-off by classification: market for electricity, low voltage, LU," ecoQuery, https://ecoquery.ecoinvent.org/3.11/cutoff/dataset/2297/impact_assessment, 2024, time period 2020–2024. IPCC 2021, climate change: total (excl. biogenic CO_2), GWP100: 0.337 kg CO_2e per kWh. Accessed 19 October 2026.

[12] G. Myhre, D. Shindell, F.-M. Bréon, W. Collins, J. Fuglestvedt, J. Huang, D. Koch, J.-F. Lamarque, D. Lee, B. Mendoza, T. Nakajima, A. Robock, G. Stephens, T. Takemura, and H. Zhang, "Anthropogenic and natural radiative forcing," in *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2013, pp. 659–740, table 8.7, GWP100 without climate–carbon feedbacks.

[13] B. Ries, "Green IT, session 2: Sustainable development," Lecture slides, MICS2-63, University of Luxembourg, Sep. 2026, slide 25 and reference [2]: ecoinvent 3.11 cut-off, transport, passenger, car, petrol, small size, EURO 5, RER, 0.291 kg CO_2e per vehicle-km at GWP100, cradle to grave.

APPENDIX A
FIGURES AND TABLES

TABLE I

LIFE CYCLE INVENTORY OF ONE RUN (SECTION III-B). MEASURED ROWS ARE THE MEDIAN OF FIVE RUNS ON `AION-0127`, EXCEPT THE RAPL ROWS (`AION-0144`); ALL ROWS READ ON 29 SEPTEMBER 2026 UNLESS STATED.

Component	Quantity	Value and unit	Source	Kind
Compute node	Walltime	368 s (range 362–372)	`sacct Elapsed`, jobs 15860349–53	measured
	Nodes, cores held	1 node, 128 cores	`sacct AllocNodes, AllocCPUS`	measured
	Memory reserved	224 000 MiB	`sacct ReqMem`	measured
	Memory touched	7 760 436 KiB (7.40 GiB)	`sacct MaxRSS`, step `.0`	measured
	Electricity, scheduler	none returned	`ConsumedEnergy(Raw)` empty	measured
	CPU-package energy per run	67 834 J (range 67 129–68 827)	RAPL, jobs 15861652–60	measured
	CPU-package power, idle	183 W (10 959 J in 60 s)	RAPL, job 15861674	measured
	Package energy above idle	1 863 J (range 1 690–1 953)	run − idle rate × step time	derived
	Electricity, 1 core active	36.7 kJ	Green Algorithms [1]	model
	Electricity, 128 cores at TDP	241 kJ	Green Algorithms [1]	model
	Hardware	2 × EPYC 7H12, 256 GB	[6], `lscpu`	looked up
	Manufacture, ecoinvent proxy	221 kg CO_2e per unit	[8], desktop without screen	looked up
	Manufacture, vendor proxy	1 768.4 kg CO_2e per unit	[9], R7525, IPCC 2013	looked up
	Service life	6 years	Section VI	assumption
	Share per run	1.94×10^{-6}	368 s / 189 345 600 s	derived
Interconnect	Bytes read from storage	≈187.7 MB + 19 MiB	`MaxDiskRead`, batch minus pipe	inferred
	Bytes written to storage	1 972 B + log	`stat` of output	measured
Storage	Input stored	187 693 955 B	`stat`, SHA-256 checked (28 Sep)	measured
	Output stored	1 972 B	`stat`	measured
Site	PUE	1.36	EU average, 2023 data [10]	declared proxy
All	Runs per year	100	Section VI	assumption

TABLE II

IMPACT CHAIN FOR ONE RUN (SECTION III-C). ENERGY VALUES ARE THE MEDIANS OF THE FIVE RUNS ON `AION-0144` (JOBS 15861652–60), EACH PAIRED WITH ITS OWN `ELAPSED`; RECOMPUTED BY `LCIA-CALCULATION.PY` ON 19 OCTOBER 2026. IPCC 2021 GWP100 UNLESS STATED.

Step	Value and unit	Source or rule	Kind
Walltime held (`Elapsed`)	364 s (range 359–368)	`sacct`, jobs 15861652–60	measured
CPU packages, Python step	67 834 J	RAPL	measured
CPU packages, rest of allocation	0.2–0.6 kJ	182.6 W idle × (`Elapsed` − step)	derived
Memory	95.4 W × 364 s = 34.7 kJ	0.3725 W/GB × 256 GB [1]	model
Node estimate	103 146 J = 0.0287 kWh	sum of the three, per run	derived
Site electricity	0.0390 kWh	× PUE 1.36 [10]	declared proxy
Grid factor	337 gCO_2e/kWh	ecoinvent 3.11, electricity low voltage, LU [11], read 19 Oct 2026	looked up
Operational	**13.1 gCO_2e**	0.0390 × 337	derived
Hardware production	913.7 kg CO_2e (IPCC 2013)	R7525 1 768.4 kg − 854.7 kg SSDs [9]	looked up
Share per run	1.92×10^{-6}	364 s / 189 345 600 s (six years)	assumption
Embodied	**1.76 gCO_2e**	913.7 kg × 1.92×10^{-6}	derived
Total	**15 gCO_2e per run**	13.1 + 1.8, two significant figures	derived

TABLE III

ENERGY SCENARIOS FOR ONE RUN (SECTION III-D), ALL ON THE 364 S MEDIAN `ELAPSED` OF `AION-0144`, WITH PUE 1.36, 337 gCO_2E/KWH AND THE CHOSEN 1.76 G OF HARDWARE PRODUCTION. THE GREEN ALGORITHMS ROWS USE 4.375 W PER CORE (280 W PER 64-CORE PROCESSOR [6]) AND 0.3725 W/GB FOR 256 GB [1].

Scenario	Node energy (kJ)	Site (kWh)	Operational (gCO_2e)	Total (gCO_2e)
CPU packages, Python step only (Section III-B)	67.8	0.0256	8.6	10.4
Chosen: packages over the allocation + memory model	**103.1**	**0.0390**	**13.1**	**14.9**
Green Algorithms, 1 core + 256 GB	36.3	0.0137	4.6	6.4
Green Algorithms, 128 cores at rated power + 256 GB	238.6	0.0901	30.4	32.1
CPU packages above idle, Python step (the program's own work)	1.9	0.0007	0.24	2.0

APPENDIX B
AI USAGE JOURNAL

AI model: Claude Opus 5.5 (`claude-opus-5-5`)
Interaction: chat, through the Claude desktop app
Prompts: 17
Input tokens: 13 418 158, mostly re-read context
Output tokens: 88 944
Energy factor: 1.03 Wh per 1000 output tokens, so 91.6 Wh
Estimated: converted to gCO_2e with the grid factor declared in Section III-C

Kinds of task, and which sections they touched: I used the chat as a guide while I set up the repository and the job, and then for general corrections and fact checking of Sections III.A and III.B and of the references. All the text was spell-checked and proofread with it. The choices of scope, boundary and measurement were mine, taken step by step with the chat.

Wrong information:

1) it split the commit hash of the job wrongly when breaking it over lines. I compared it with the repository and corrected it.
2) it described the end-of-life treatment of the cut-off model wrongly. I rewrote the sentence.
3) some reference details were wrong: the release year of ecoinvent 3.11 and the data year of the EU PUE. I checked each reference at its source.